

Maximilian Bershtman

M.Sc. Student, Electrical Engineering · Artificial Intelligence / Machine Learning / Deep Learning

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SUMMARY

M.Sc. student in Electrical Engineering at the Technion, focusing on machine learning, deep learning, and artificial intelligence. Currently researching tabular anomaly detection with prior-data fitted networks under Prof. Niv Cohen, with a paper in preparation for ICLR 2027 that will form the basis of my thesis. B.Sc. in Information Systems Engineering, with earlier research in graph neural networks and applied data science.

EDUCATION & RESEARCH

M.Sc., Electrical Engineering · Technion - Israel Institute of Technology

Oct 2025 - Present

Specialization in artificial intelligence, machine learning, and computer vision. Coursework: Deep Learning, Modern Computer Vision, Computer Vision Laboratory, Machine Learning, Computational Methods in Optimization.

Thesis research · Tabular Anomaly Detection with Prior-Data Fitted Networks - Advisor: Prof. Niv Cohen

- Adapting TabPFN, a pretrained tabular foundation model, into an anomaly detector that stays reliable when the reference data it learns from is itself contaminated with anomalies.
- Developed an embedding-space scoring method and a compactness fine-tuning scheme, ensembled with classical detectors. The resulting method is **state of the art in tabular anomaly detection**, ranking first among eighteen classical and deep detectors on the field's standard benchmark suite.
- Paper in preparation for submission to ICLR 2027; this work will form the basis of my M.Sc. thesis.

Research project · Semantic Probing of Iterative 3D Reconstruction Models - Advisor: Yoni Kasten, NVIDIA

- Investigated whether a transformer-based 3D scene reconstruction model implicitly learns object-level semantics, by training a lightweight probing head on its frozen latent representations.
- Showed the model can segment objects from camera viewpoints it was never shown and generalize to unseen object instances, indicating a 3D-consistent semantic representation.

B.Sc., Information Systems Engineering · Technion – Israel Institute of Technology

Oct 2021 – Oct 2025

GPA: 85.7. Focus areas: mathematics, statistics, computer science, data science, machine learning, deep learning, artificial intelligence.

Selected coursework: Machine Learning 1 & 2, Probabilistic Graphical Models, Introduction to Causal Inference, Foundations and Applications of Artificial Intelligence, Practical Principles for Image Processing, Data Structures and Algorithms, Database Management, Distributed Information Systems, Statistics.

PUBLICATIONS & REPORTS

- Adapting Prior-Data Fitted Networks for Tabular Anomaly Detection Under Uncertainty - in preparation for ICLR 2027, with Prof. Niv Cohen.
- Semantic Probing of Iterative 3D Reconstruction Models: Extending Cut3r - technical report, with Yoni Kasten (NVIDIA), 2025.

PROFESSIONAL EXPERIENCE

Data Analysis Research Assistant · Technion - Faculty of Data and Decision Sciences Jan 2024 – Jan 2025

- Conducted data analysis on research datasets - cleaning, preprocessing, and feature engineering in Python (Pandas, NumPy, SciPy).
- Applied the OpenAI API for research automation and natural language processing tasks.
- Supported meta-analysis projects by aggregating and synthesizing findings across multiple studies.

Deep Learning Research Assistant · Technion - Faculty of Chemical Engineering Mar 2023 – Mar 2024

- Developed Graph Neural Network models with PyTorch Geometric to predict micro-level geometric changes in molecular chemistry simulations.
- Improved model performance through hyperparameter tuning, data augmentation, and architecture refinement.

System Administrator · Israel Defense Forces

SKILLS

Machine Learning & AI: Deep Learning, Computer Vision, Anomaly Detection, Graph Neural Networks (GNN), Foundation Models, Transformers, NLP, PyTorch, PyTorch Geometric, scikit-learn.

Programming & Tools: Python, SQL, Java, Object-Oriented Programming, Git, Microsoft Office.

Data & Research: Data Analysis, Statistics, Optimization, Experimental Design, Benchmarking, Meta-Analysis.

Languages: Hebrew (native), Russian (native), English (proficient).